

STUDENT NAME and ID: _____

1. (3+3+4 marks) Consider a **linear regression** problem involving **a single input feature**, denoted as $\phi(x)$. You are provided with a set of n training examples $\{(\phi(x^{(1)}), y^{(1)}), (\phi(x^{(2)}), y^{(2)}), \dots, (\phi(x^{(n)}), y^{(n)})\}$. The superscript indicates the respective training indices.
 - (a) Finish the **linear predictor model** equation: $f_w(x) = \hat{y} = (??)$. Provide an explanation for the equation
 - (b) Formulate the expression for the **residual** and a **loss function** of your choice. Elaborate on the rationale behind your chosen loss function.
 - (c) Identify whether the problem falls under **continuous or discrete optimization**. Articulate the optimization problem, including its formulation. Define the solution to the optimization problem and clarify its role in the overall process—whether it pertains to training or testing. Describe your approach to solving this optimization problem.
2. (4+3+3 marks) Let's now consider a **linear regression** problem characterized by **multiple input features (involving d features)** $\vec{\phi}(x) = [\phi_1(x), \phi_2(x), \dots, \phi_d(x)]$. You are given n training examples $\{(\vec{\phi}(x^{(1)}), y^{(1)}), (\vec{\phi}(x^{(2)}), y^{(2)}), \dots, (\vec{\phi}(x^{(n)}), y^{(n)})\}$. Repeat parts (a-c) from Question 1 for Question 2.

Answers

1. (a) $f_w(x) = \hat{y} = w_1\phi(x)$. Explanation: **open-ended** - any reasonable explanation will be accepted. One example explanation: As we have one one input feature, the number of weight parameters will be 1
- (b) Residual = $(\hat{y} - y)$ or $(f_w(x) - y)$. Loss function, squared loss $J(w_1) = \sum_{all-training-data} residual^2 = \sum_{i=1}^n (\hat{y}^{(i)} - y^{(i)})^2 = \sum_{i=1}^n (f_w(x^{(i)}) - y^{(i)})^2 = \sum_{i=1}^n (w_1\phi(x^{(i)}) - y^{(i)})^2$. Explanation: **open-ended**.
- (c) Continuous. Optimization problem:

$$\min_{w_1} J(w_1) = \min_{w_1} \sum_{i=1}^n (w_1\phi(x^{(i)}) - y^{(i)})^2 \quad (1)$$

Solution of the optimization problem - optimum weight parameter (or model parameter) w_1 . Part of training process. How to solve - many options, one option is using gradient descent.

2. (a) $f_w(x) = \hat{y} = w_1\phi_1(x) + w_2\phi_2(x) + \dots + w_d\phi_d(x) = \vec{w} \cdot \vec{\phi}(x)$. Here, $\vec{\phi}(x) = [\phi_1(x), \phi_2(x), \dots, \phi_d(x)]$ and $\vec{w} = [w_1, w_2, \dots, w_d]$. Explanation: [open-ended](#).

(b) Residual = $(\hat{y} - y)$ or $(f_w(x) - y)$. Loss function, squared loss $J(\vec{w}) = \sum_{all-training-data} \text{residual}^2 = \sum_{i=1}^n (\hat{y}^{(i)} - y^{(i)})^2 = \sum_{i=1}^n (f_w(x^{(i)}) - y^{(i)})^2 = \sum_{i=1}^n (\vec{w} \cdot \vec{\phi}(x^{(i)}) - y^{(i)})^2$. Explanation: [open-ended](#).

(c) Continuous. Optimization problem:

$$\min_{\vec{w}} J(\vec{w}) = \min_{\vec{w}} \sum_{i=1}^n (\vec{w} \cdot \vec{\phi}(x^{(i)}) - y^{(i)})^2 \quad (2)$$

Solution of the optimization problem - optimum weight parameters (or model parameters) $w_1, w_2, \dots, w_d$ or weight vector (or model parameter vector) $\vec{w}$. Part or training process. How to solve - many options, one option is using gradient descent.